Before its fourth collision with a plate, the ball has kinetic energy

From ,

Its speed just before the fourth collision is

Its speed and kinetic energy just after the fourth collision are

Continuing likewise, its kinetic energies just before and after collision are

Thus

For inelastic collisions, ; and grow with the collision count. As ,

The maximum kinetic energy is the limiting value just before a collision, so

After a sufficiently long time, as , the speeds just after one collision and just before the next are